

Mohammad Mobassir Ameen

I am a particle physicist specializing in experimental high-energy physics. I am currently a PhD student at IIT Madras, actively engaged in research within the CMS collaboration, with a focus on Higgs physics. As part of my research experience, I have worked on the performance of monolithic pixel sensors under X-rays irradiation, contributing to detector development and characterization studies.

Education

- 2019–present **PhD, Physics**, *Indian Institute of Technology*, Madras, India.
Thesis Title: Studies of Higgs Boson Production in Association with Top Quarks in Multilepton Final States at CMS Experiment and Performance of Monolithic Pixel Sensor under X-ray Irradiation
Advisor: Prof. Prafulla Kumar Behera
Status: Thesis submitted; defense expected by the end of August 2026.
CGPA : Course Work: 8.0/10
- 2016–2018 : **Master of Science in Physics**, *Aligarh Muslim University*, Aligarh (U.P.), India.
CGPA : 7.29/10
- 2012–2015 : **Bachelor of Science (Hons.) in Physics**, *Aligarh Muslim University*, Aligarh (U.P.), India.
MARKS : 920/1500
- 2010–2012 : **Intermediate**, *Dr Ashok Kumar Smarak Inter College*, Akbarpur, Ambedkar Nagar (U.P.), India.
Subjects: Mathematics, Chemistry, Physics, English, General Hindi
Percentage : 73.2 %
- 2010 : **High School**, *Dr Ashok Kumar Smarak Inter College*, Akbarpur, Ambedkar Nagar (U.P.), India.
Subjects: Mathematics, Social Science, Science, English, Urdu, Hindi
Percentage : 66.66 %

Publications

Journal Article(Accepted)

- 2020–present: **M. M. Ameen**, as a member of the CMS experiment collaboration, I have contributed to a large number of published papers. According to the CMS authorship policy, a researcher qualifies as an author by actively participating in the data-taking operations and contributing to essential service tasks. The data-taking process is a continuous, large-scale effort that requires significant manpower to operate and monitor the CMS detector daily. As a result, each CMS publication typically includes around a thousand authors, reflecting the collective contribution of the collaboration. During my time in the collaboration, I have fulfilled these requirements through my service work. In particular, I served as an L3 PdmV validation manager, where I was involved in Physics Data and Monte Carlo validation. In addition, I was part of the Higgs physics group as a Monte Carlo contact person, responsible for coordinating and validating the simulation of MC samples relevant to Higgs physics analyses. Through these contributions, I have been included as a co-author on more than 200+ CMS publications, 7,500+ citations (as of July 2026). The complete list of these publications can be accessed by clicking on the following link: [INSPIREP](#) or [ORCID](#).

A few of the analyses in which I have made my significant indirect contributions (by simulating the MC samples) are published and listed below:

- 2026 **The CMS Collaboration.** Search for nonresonant HH production in the $b\bar{b}\tau\tau$ final state in proton-proton collisions at $\sqrt{s} = 13.6$ TeV. Technical report, CERN, Geneva, 2026.
- 2026 **The CMS Collaboration.** Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV. *JHEP*, volume 08, page 019, 2026.
- 2026 **The CMS Collaboration.** Observation of tWZ Production at the CMS Experiment. *Phys. Rev. Lett.*, volume 136, page 081802, 2026.
- 2026 **The CMS Collaboration.** Improved results on Higgs boson pair production in the $4b$ final state. 4 2026.
- 2025 **The CMS Collaboration.** Measurements of the Higgs boson production cross section in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13.6$ TeV. *JHEP*, volume 05, page 079, 2025.
- 2025 **The CMS Collaboration.** Measurement of WWZ and ZH Production Cross Sections at $\sqrt{s} = 13$ and 13.6 TeV. *Phys. Rev. Lett.*, volume 135, page 091802, 2025.
- 2025 **The CMS Collaboration.** Measurement of the inclusive WZ production cross section in pp collisions at $\sqrt{s} = 13.6$ TeV. *JHEP*, volume 04, page 115, 2025.
- 2025 **The CMS Collaboration.** Measurement of inclusive and differential cross sections for W^+W^- production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV. *Phys. Lett. B*, volume 861, page 139231, 2025.

Recent Journal Published Article (IITM Group)

- 2026 Mohammad Mobassir Ameen, Ganapati Dash, Anushree Vijay, Theertha Chembakan, and Prafulla Kumar Behera. Performance of monolithic cmos pixel sensors under x-rays. *Physics*, volume 8, 2026.

Ongoing Journal Article

- 2026 **The CMS Collaboration,** Study of Higgs boson production in association with top quarks in multilepton final states using the Run 3 dataset.
Internal Note: CMS AN-25-160, CMS HIG-26-008 (**Status:** Group Review)

Personal Contributions to few other CMS Analyses

- 2026 **The CMS Collaboration,** Search for new physics in top quark production with additional leptons in proton-proton collisions from Run 2 at $\sqrt{s} = 13$ TeV and Run 3 at $\sqrt{s} = 13.6$ TeV using Effective Field Theory, My contribution: Derived the dilepton and trilepton trigger-efficiency scale factors used as corrections in this analysis.
Internal Note: CMS AN-24-174, CMS TOP-26-006 (**Status:** Group Review)
- 2025 **The CMS Collaboration,** Measurement of the inclusive ttW cross section with Run 3 data, My contribution: Derived the dilepton and trilepton trigger-efficiency scale factors used as corrections in this analysis.
Internal Note: CMS AN-24-239, CMS TOP-25-014 (**Status:** Preapproved)
Trigger SFs Approved by: **Muon POG** and **EGamma POG**.

In Conference Proceedings

- 2025 **Mohammad Mobassir Ameen.** Measurement of the ttH Production Cross-Section in Multi-leptonic Final States in pp Collisions at a Centre-of-Mass Energy of 13 TeV with the CMS Detector. In *Proceedings of the 17th International Conference on Interconnections Between Particle Physics and Cosmology*, pages 297–301. Springer Nature Singapore, 2025.
- 2024 **Mohammad Mobassir Ameen.** Observation of the Decays $B^0 \rightarrow \psi(2S)K_S^0\pi^+\pi^-$ and $B_S^0 \rightarrow \psi(2S)K_S^0$ at $\sqrt{s} = 13$ TeV in the CMS Experiment. In *Proceedings of the XXV DAE-BRNS High Energy Physics (HEP) Symposium 2022*, pages 725–726. Springer Nature Singapore, 2024.

Research Experience

PhD Research Work

2023 – present **Higgs boson production in association with top quarks in multilepton final states.**

In this analysis, we studied of Higgs boson production rate in association with one (tH) and two (ttH) top quarks in final states with electrons, muons, and tau leptons. The analysis is based on proton–proton collision data at 13.6 TeV collected during 2022–2023 by the CMS experiment, corresponding to an integrated luminosity of 62 fb^{-1} . Advanced machine learning techniques, including transformer-based neural networks, are used to enhance signal-background discrimination and to separate tH and ttH processes. The production rates are measured relative to SM predictions, and constraints are placed on the top quark Yukawa coupling (y_t) under Standard Model assumptions. A blinded statistical framework is employed to extract signal strength modifiers and determine allowed parameter regions at 95% confidence level.

Personal Contributions:

- Generated signal and background Monte Carlo samples using MadGraph and Pythia.
- Computed trigger efficiencies and derived corresponding scale factors, approved by the Muon and EGM POGs (Adopted in the TOP-25-014 and TOP-26-006 (Run-3 extension of the TOP-22-006 CMS analysis) CMS analysis).
- Developed a Higgs jet tagger for improved signal identification.
- Designed and trained transformer-based machine learning models for signal extraction.
- Implemented the full analysis workflow within the CMGRDF framework.
- Incorporated relevant systematic uncertainties into the analysis.
- Performed likelihood scans and extracted signal strength using the Higgs Combine tool.

2023 – present **Search for CP violation in association with top quarks in multilepton final state.**

Investigation of the CP structure of the Higgs boson coupling to top quarks as an extension of the tH and ttH multilepton analysis. This study builds upon the same analysis framework and dataset, incorporating additional CP-sensitive observables. Machine learning techniques are further adapted to distinguish between CP-even and CP-odd scenarios in final states with multiple leptons. A dedicated statistical interpretation is developed to extract confidence regions for CP-sensitive coupling parameters and to probe potential CP-violating effects.

Personal Contributions:

- Extended the full analysis framework developed for ttH/tH measurements to probe CP properties.
- Implemented CP-sensitive observables and corresponding systematic uncertainties.

CADI: **CMS HIG-26-008**

2025 – 2026 **CMOS monolithic pixel sensor under X-rays.**

Study of monolithic CMOS pixel sensors under X-ray irradiation to evaluate their performance for high-energy physics tracking detectors. Performed detailed characterization of threshold and noise behaviour as a function of front-end DAC tuning parameters.

Personal Contributions:

- Led conceptualization, methodology design, data analysis, and validation
- Performed data collection, curation, and investigation
- Developed visualizations; and contributed to manuscript writing.

This work: **Published in Journal Physics**

CMS Service Work

2020 – 2022 **Validation of pre-release and production release of CMS Software (CMSSW).**

In the Physics Data and Monte Carlo Validation (PdmV) group, served as an L3 manager, I was responsible for validating the developments for each pre-release and production release of CMSSW. The validation chain starts from defining and requesting release validation (relval) samples production. Once samples are ready, relval comparison plots (relmon) are prepared comparing target and reference. Then standard validation campaign should be open at validation database (valdb), providing all related information (for e.g., goal of validation campaign, relval samples, relmon and Global tag difference between target and reference etc.) to all validators of Physics Object Group (POG) of the CMS Collaboration. Then, validation reports will come from all the groups in valdb. In the end, we summarize the results of the validation campaign at PPD general meeting. In this group, I have been working for two years with more than 20 production software.

Coordinators : **Bugra Billin, Kirill Skovpen**

2023 – 2025 **Production Of Monte Carlo Simulated Samples For The Higgs Physics Analysis Group (HIG PAG).**

As a ttH and di-Higgs MC contact, which is an important role for the HIG PAG, especially with regard to ongoing Run-3 production. In which my responsibilities and activities are to coordinate with HIG PAG analysis teams for preparation, validation & submission of gridpacks of MC samples (signal and background), i.e. Act as a "bridge" between analysis proponents and the overall HIG MC contacts, who in turn are in direct communication with GEN & PPD/PdmV & HIG PAG conveners. After receiving the request from the analyser, the MC has to prepare, validate & submit the tickets with the Monte-Carlo Management (McM) system. These are reviewed on a weekly basis during the MccM Meeting. We must attend all weekly meetings of the Monte Carlo Coordination Meeting (MccM) and the GENerator (GEN) to stay in touch with developments and follow-up on submission and monitoring, in coordination with HIG MC contacts to ensure that sample production is proceeding smoothly. Till the end of the task, I have dealt with more than hundred requested samples which is used in the Higgs physics analysis.

Main HIG MC Contacts: **Alexandros Attikis, Mattia Lizzo, Yihui Lai, Braden Kronheim**

Talks and Presentations

- 14 - 18 Oct 2024 "Measurement of the ttH production cross-section in multi-leptonic final states in pp collisions at a centre-of-mass energy of 13 TeV with the CMS detector" at XVII International Conference on Interconnections between Particle Physics and Cosmology: PPC 2024, IIT Hyderabad, India.
- 09 - 18, Oct 2023 "ttH and tH production in the multilepton channel using Run-III dataset at CMS" in the Physics Object & Data Analysis School, at DESY, Hamburg, Germany.
- 12-16, Dec 2022 "Observation of the decay $B^0 \rightarrow \Psi(2S)K_s\pi^+\pi^+$ and $B_s^0 \rightarrow \Psi(2S)K_s^0$ at 13 TeV pp collision in the CMS experiment" in the XXV DAE-BRNS High Energy Physics Symposium, IISER Mohali, India.
- 29-30, July 2022 "Study of the B_c decays to $B_s\pi$ at the CMS detector", In-House Symposium, Department of Physics, IIT Madras, India.

Schools, Conferences, Workshops, Seminars

- 20–30 July 2026 Summer School on High Energy Physics (Virtual Zoom)
- 5–9 Jan 2026 HSF-India Scientific Computing Workshop at IIT Madras, India
- 5-9 Feb 2024 CMS Week at CERN, Geneva, Switzerland
- 12-13 Feb 2024 CMS Physics Days: Ideas of Physics Analysis with '2022' and '2023' Data at CERN, Geneva, Switzerland
- 29 Jan-2 Feb 2024 6th Inter-experiment Machine Learning Workshop at CERN, Geneva, Switzerland
- 14 - 18 Oct 2024 XVII International Conference on Interconnections between Particle Physics and Cosmology: PPC, IIT Hyderabad, India
- 4-14 Jan 2022 CMSDAS by the LHC Physics Center at Fermilab, USA (Virtually)
- 12-25 Feb, 2023 XV ICFA School on Instrumentation in Elementary Particle Physics at TIFR Mumbai, India
- 1-5 May, 2023 HSF-India Training event at TIFR, Mumbai, India
- 15-16 Nov 2022 Latino Framework Workshop at CERN [Online]

- 29-30, July 2022 In-House Symposium, Department of Physics, IIT Madras, India
- 25-31 May 2021 Geant4 Beginners Course @ CERN [virtual], CERN (Geneva)
- 16-24 Dec 2019 Cosmic Neutrino Observation at Ultra-High Energy, IIT Kanpur, India

Scholastic Achievements

- 2016 IIT JAM (Joint Admission Test for Masters)
- 2018 Graduate Aptitude Test in Engineering (GATE)
- Dec 2018 CSIR-NET

Fellowships & Awards

- 2019–2024 **Half-time Research Assistantship (HTRA) Fellowship**, as a PhD research scholar in Indian Institute of Technology Madras.
- 2023 Awarded **International Immersion Experience (IIE) Travel Award, IITM** to attend any international university.
- 2023–2024 Awarded **Advance Research Opportunities Program (AROP)** to visit **RWTH Aachen University, Germany** for the short research from 01st October 2023 to 31th March 2024.

Academic Achievements & Recognitions

- 2023 **Best Poster Award** in the **Physics Object & Data Analysis School** during 9-18 October 2023 at **DESY, Hamburg, Germany**.
- 2025 **Breakthrough Prize in Fundamental Physics** (shared with colleagues in the CMS Collaboration).

Memberships

- 2020–present Associated member of the Compact Muon Solenoid (CMS) experiment at the Large Hadron Collider (LHC) accelerator at the European Organization for Nuclear Research (CERN)
- 2016–2018 Member of the **Aligarh Muslim University Society of Physics (AMUSoP)** : A student body with a mentoring group as faculty members, organizes various academic activities like inviting talks of the faculty members of the university and the outside of the university, group discussions, workshops etc.

Languages

Hindi, Urdu, English

Computer skills

- Programming Languages Python, PyTorch, keras, C, C++, Bash/Shell Scripting
- HEP Tools & Frameworks CMSSW (CMS Software), Root, RooFit, Git and Github, GitLab, CMGTools based on RDataFrame, Higgs Combine tool
- Word Processing Competent with Microsoft Office, Latex, Overleaf
- Machine Learning & AI Binary Classification Algorithm for e.g., expertise in BDT, XGBoost and Multiclassification Algorithm like DNN, Transformer

Position of Responsibility

- 2023–2026 **ttH+HH Monte Carlo Contact in HIG Monte Carlo (MC) team**, CMS Collaboration, CERN.
- 2020–2022 **L3 Validation manager in PdmV (Physics Data and Monte Carlo Validation)**, CMS Collaboration, CERN.
- 2019 **L1 trigger shift as online trigger shifter for Beam Test and CRAFT (Cosmic Run and Full Tesla)**.

Teaching Assistantship

- Fall, 2023 : **EP3291: Engineering Physics Lab IV**, *Department of Physics*, IIT Madras.
- Spring, 2022 : **PH1010: PHYSICS I : Classical Mechanics**, *Department of Physics*, IIT Madras.
- Fall, 2022 : **PH1020: PHYSICS II : Electricity & Magnetism**, *Department of Physics*, IIT Madras.

Mentoring

- Jan-May, 2026 : **Mentored a M.Sc. Physics (IITM) on the project**, *Machine Learning based Higgs-Jet Tagging in ttH multilepton Analysis*.
- May-July, 2026 : **Mentored three M.Sc. Physics intern students from University of Madras**, *Guided them in high-energy physics data analysis software tools, ROOT/Python, and machine learning applications*.

Hobbies

Travelling (exploring new cities/countries culture and their food), Interacting with the researcher to learn more

Referees

Dr. Prafulla Kumar Behera

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Dr. Najimuddin Khan

Assistant Professor, Department of Physics

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Dr. Meng Xiao

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Dr. Pallabi Das

Experimental Research Fellow

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